Weather Station Logger Application

Version 2.5

Installation

Run the install file WsLoggerInstaller.exe, by default this will place the WsLogger application files in the folder "C:\Program Files (x86)\WsDataLogger", an alternative location can be chosen. On completion of the installation the following files should be present in the chosen folder:

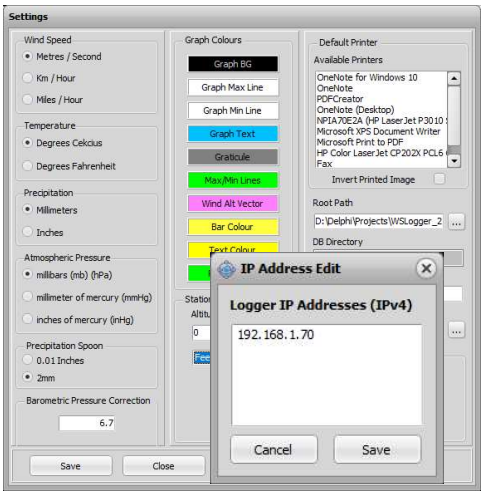
WsLogger.Exe	Weather Data viewer app
WsIsam.dll	DLL holding DBIsam data table routines
borlndmm.dll	Borland Memory Manager

The first time the application is run

- You will be prompted for the IP address of the Logger Unit, enter this in IPv4 form (four octets).
- The Ini file for the application will be created and the application will run.
- A sub folder (DB) in the install location will be created.
- Empty data files will be created in the DB folder.

Configuration

The configuration page can be found under the Settings menu option, click on this and the settings window will open.



The following settings **critically** affect operation of the application:

- Root Data Path
- DB Directory
- Database table name
- Selected Host IP – This MUST be the IP address of the computer on which the WsLogger application is running. If you click on the ellipsis button the IP Address Edit box will open. If there is more than one address shown then select the correct one and press "Save".

Other settings:

- Units for wind, rainfall, temperature and barometric pressure
- Precipitation measuring spoon type
- Weather station altitude in selected units (barometric pressure correction)
- Logger pressure correction relative to a Davis console.
- Graph colours
- Default Printer
- Auto import settings
    - Check "At Startup" if you want new data loaded whenever you run the app.
    - Check "Every Hour" if you want new data to be automatically loaded once an hour.

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Instructions

Tabs at the bottom left of the application panel.

These instructions presume there to be valid data within the database!

Analyse Tab

On the calendar shown select a date. The data display period can be chosen from one day to 24 months, start with one day. The data for the selected date and period can now be viewed by pressing one of the activity buttons on the toolbar.

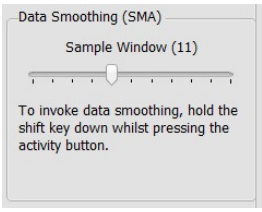
Data display activity's

Wind Rose – This displays wind direction, speed information and period as a Wind Rose

Wind Graph – This displays wind direction as a graph with the Y axis in degrees. As wind direction is chaotic in the short term, data smoothing can be applied. The wind direction graph reflects the directional position of the vane on the weather station irrespective of whether or not the wind is blowing.

Wind Polar – This displays the relative wind speeds and direction for four, six-hour periods. The width of each angular segment is 10 degrees.

Wind Speed Adv	Averaged wind speed. Wind speed is chaotic so data smoothing can be applied.
Wind Speed Peak	The peak wind speed measured in ten-minute periods, a gust measurement.
Temperature	Temperature graph with maximum and minimum lines.
Humidity	Humidity graph with maximum and minimum lines.
Rainfall Text	Rainfall shown in text form.
Rainfall Graph	Rainfall shown in bar graph form.
Rainfall Rate	Rainfall rate in text form
Barometer	Barometric pressure graph with maximum and minimum lines.
RSSI	Received signal strength indication (for the weather station)

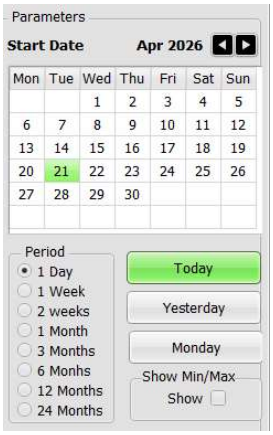


Data smoothing can be applied to any of the above. In the Data Smoothing box select the amount of smoothing (11 is optimal). Hold down the shift key and press the desired activity button the toolbar. The smoothing is SMA (Simple Moving Average).

Graph Period

Period This the period that the graph encompasses. 1 Day to 2 years.

Graph Date This can be selected to the required date. In addition there are buttons to move the selected date to Today, Yesterday or Monday of the current week.



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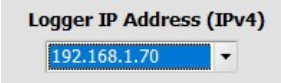
Weather Station Tab

Logger IP Address	Drop the Logger IP addresses box down and select the IP address of your logger device. Multiple loggers are permitted but only one can be selected at a time. 
Get directory from logger	This loads a list of data files that are present on the logger device.
Import Checked files	This imports the data from any files checked in the directory list.
Erase Checked Files	This will erase any files that have been checked in the directory list.
Auto import new data	This will collect any new data from the logger and automatically import it.
Logger Status	This will collect and display the status of the logger. This may take up to ten seconds.
Logger System Log	This will download and display the loggers internal log.

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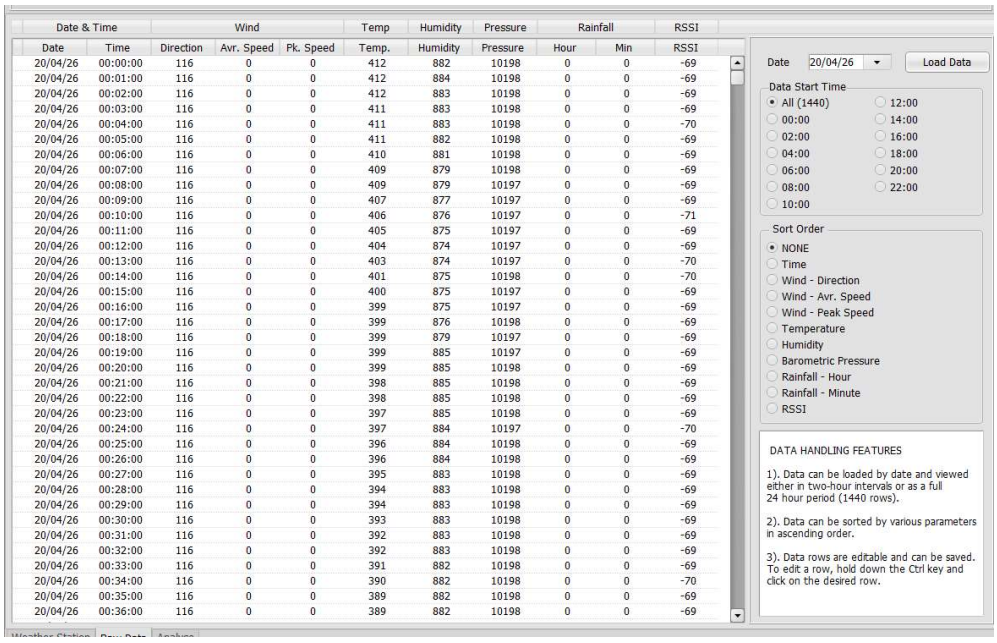
RAW Data Tab

This will display the data for the selected date.

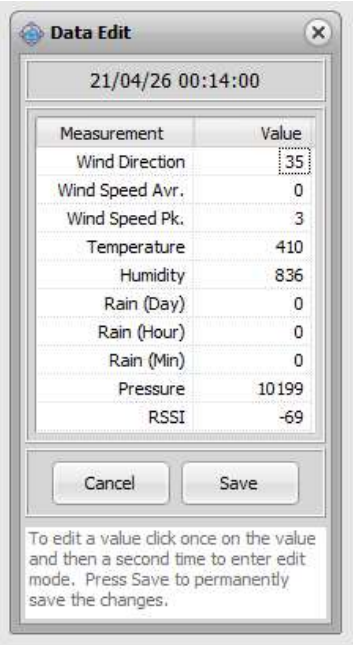
The data can be displayed as twelve two hour blocks or as one 24 hour block.

The data columns are Normally in time order But can be ordered by any of the sorting Parameters.

Data rows are editable and can be saved. To edit a row, hold down the Ctrl key and click on the desired row. On completion of the edit press save.



Date & Time		Wind			Temp	Humidity	Pressure	Rainfall		RSSI
Date	Time	Direction	Avr. Speed	Pk. Speed	Temp.	Humidity	Pressure	Hour	Min	RSSI
20/04/26	00:00:00	116	0	0	412	882	10198	0	0	-69
20/04/26	00:01:00	116	0	0	412	884	10198	0	0	-69
20/04/26	00:02:00	116	0	0	412	883	10198	0	0	-69
20/04/26	00:03:00	116	0	0	411	883	10198	0	0	-69
20/04/26	00:04:00	116	0	0	411	883	10198	0	0	-70
20/04/26	00:05:00	116	0	0	411	882	10198	0	0	-69
20/04/26	00:06:00	116	0	0	410	881	10198	0	0	-69
20/04/26	00:07:00	116	0	0	409	879	10198	0	0	-69
20/04/26	00:08:00	116	0	0	409	879	10197	0	0	-69
20/04/26	00:09:00	116	0	0	407	877	10197	0	0	-69
20/04/26	00:10:00	116	0	0	406	876	10197	0	0	-71
20/04/26	00:11:00	116	0	0	405	875	10197	0	0	-69
20/04/26	00:12:00	116	0	0	404	874	10197	0	0	-69
20/04/26	00:13:00	116	0	0	403	874	10197	0	0	-70
20/04/26	00:14:00	116	0	0	401	875	10198	0	0	-70
20/04/26	00:15:00	116	0	0	400	875	10197	0	0	-69
20/04/26	00:16:00	116	0	0	399	875	10197	0	0	-69
20/04/26	00:17:00	116	0	0	399	876	10198	0	0	-69
20/04/26	00:18:00	116	0	0	399	879	10197	0	0	-69
20/04/26	00:19:00	116	0	0	399	885	10197	0	0	-69
20/04/26	00:20:00	116	0	0	399	885	10198	0	0	-69
20/04/26	00:21:00	116	0	0	398	885	10198	0	0	-69
20/04/26	00:22:00	116	0	0	398	885	10198	0	0	-69
20/04/26	00:23:00	116	0	0	397	885	10198	0	0	-69
20/04/26	00:24:00	116	0	0	397	884	10197	0	0	-70
20/04/26	00:25:00	116	0	0	396	884	10198	0	0	-69
20/04/26	00:26:00	116	0	0	396	884	10198	0	0	-69
20/04/26	00:27:00	116	0	0	395	883	10198	0	0	-69
20/04/26	00:28:00	116	0	0	394	883	10198	0	0	-69
20/04/26	00:29:00	116	0	0	394	883	10198	0	0	-69
20/04/26	00:30:00	116	0	0	393	883	10198	0	0	-69
20/04/26	00:31:00	116	0	0	392	883	10198	0	0	-69
20/04/26	00:32:00	116	0	0	392	883	10198	0	0	-69
20/04/26	00:33:00	116	0	0	391	882	10198	0	0	-69
20/04/26	00:34:00	116	0	0	390	882	10198	0	0	-70
20/04/26	00:35:00	116	0	0	389	882	10198	0	0	-69
20/04/26	00:36:00	116	0	0	389	882	10198	0	0	-69



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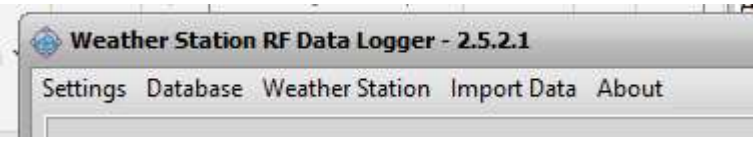
Menu options



Settings

- Edit Settings — This will take you into the settings page, most important settings are here.
- Printer setup — Allows you to configure the printer in use.

Database

- Empty Database — This will completely empty the database, use with caution.

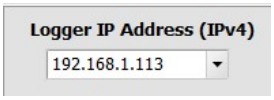
Weather Station

- IP Addresses

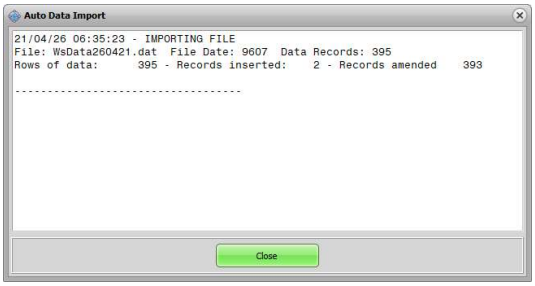
  This opens the Logger IP Address window. The window allows you to enter one or more IP address for logger units. Place each address on a different line. The active logger is selected In the Weather Station tab.

- Logger

  Restart Logger — This will restart the logger when required.



The logger automatically restarts at 03:00 every day, this is shown in its log.

- Import Data

  This will immediately start the process of importing new data. A log showing the progress will be shown. The addition of data starts with the current date and works backwards until what is in the database matches what is on the logger and then stops.